There are three components that vary in the gas law equations: pressure, temperature, and volume. When one of these components stays constant, the three gas laws predict the relationship between the other two. When you are faced with a calculation problem, you must be able to deduce from the question which components are varying, which one stays the same, and decide which formula to use.

The mnemonic **P**aid **TV C**an **B**e **G**ood can help you figure out which formula you need. To use the mnemonic, write the letter PTV and CBG so that they appear like this:

P	**T**	**V**
C	**B**	**G**

The letters P, T, and V stand for pressure, temperature and volume. The letters C, B, and G represent Charles', Boyle's, and Gay-Lussac's laws. The letter directly above each of letters in the bottom row tells you which variable is constant in that law.

For example, Charles' law ($V_1T_2 = V_2T_1$), only deals with temperature and volume. This means that pressure is held constant and ignored. If you look at the chart, you will see that the P above the C for Charles' law indicates that pressure is constant and ignored, and temperature and volume are the variables used.

P	**T**	**V**
C	**B**	**G**

Pressure is held constant in Charles' law

Likewise, in Boyle's law ($P_1V_1=P_2V_2$), temperature is ignored and only pressure and volume are a part of the formula.

P	**T**	**V**
C	**B**	**G**

Temperature is held constant in Boyle's law

And finally, in Gay-Lussac's law ($P_1T_2 = P_2T_1$), volume is ignored and only pressure and temperature are part of the equation.

P	**T**	**V**
C	**B**	**G**

Volume is held constant in Gay-Lussac's's law

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INTRODUCTION

Before we get into the individual gas laws themselves, it is important to recognize that when dealing with any of the gas laws, temperature values ***must*** be expressed in degrees Kelvin. If you are unsure how to convert temperatures from Celsius or Fahrenheit into Kelvin, then see the article in this software entitled 'Temperature Conversions' to learn how. Also, it doesn't matter what units the volume or pressure are in, as long as they are the same on both sides of the equation. If a question gives you the initial pressure in atmospheres, but then asks for an answer in mmHg, you must calculate your answer initially in atmospheres, and then convert the answer into mmHg.

CHARLES' LAW

$$V_1T_2 = V_2T_1$$

Charles' law describes the relationship between volume and temperature. The basic question that can be answered with Charles' law is: if I start with a certain volume of a gas at one temperature, then what volume will the gas occupy if I change the temperature? Everything else related to Charles' law is just a variation of this concept. Such as, if a gas occupies a certain volume at a given temperature, what will the temperature of the gas be if I compress it into a smaller volume?

As an example, let's imagine that a gas occupies a volume of 1 liter at a temperature of 150 degrees Kelvin. What volume will it occupy at a temperature of 200 degrees Kelvin?

Our initial volume is V1 and our initial temperature is T1. It is the temperature (T2) that changes and we must solve for V2 to find out how the volume has changed:

$$V_1T_2 = V_2T_1$$

$$(1L)(200K) = (V_2)(150K)$$

$$(V_2) = (1L)(200K)/(150K)$$

$$V_2 = 1.33\ L$$

So, if we have a gas with a volume of 1 liter at 150 degrees Kelvin, and then heat it to 200 degrees Kelvin, it will expand to 1.33 liters. Always check your logic when you are done to make sure that your answer makes sense: as we heated the gas, it expanded to a larger volume. Gases expand when they are heated, so we know that this answer makes sense.

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Now, let's imagine an example of a gas that comprises a volume of 6.5 liters at a temperature of 300 degrees Kelvin. If we reduce the temperature to 210 degrees Kelvin, what volume would it occupy? In this question, we are cooling the gas, which causes it to contract to a smaller volume. We should expect, then, that our answer should be a smaller volume than the one with which we started.

$$V_1T_2 = V_2T_1$$

$$(6.5L)(210K) = (V_2)(300K)$$

$$(V_2) = (6.5L)(210K)/(300K)$$

$$V_2 = 4.55\ L$$

As we expected, as we decreased the temperature, the gas contracted from a volume of 6.5 liters, to a smaller volume of 4.55 liters. Charles' law demonstrates that temperature and volume have a direct relationship, which means that the values always move in the same direction. As the temperature decreases, the volume also decreases and vice versa.

BOYLE'S LAW

Boyle's law describes the relationship between pressure and volume. A typical question that requires Boyle's law to solve is: if a gas occupies a volume of 2 liters at a pressure of 2 atmospheres (atm), what volume will it occupy at a pressure of 4 atmospheres? In this problem, P_1 = 2 atm, V_1 = 2 liters, P_2 = 4 atm, and we must solve for V_2.

$$P_1V_1 = P_2V_2$$

$$(2\ atm)(2\ L) = (4\ atm)(V_2)$$

$$V_2 = (2\ atm)(2\ L)/(4\ atm)$$

$$V_2 = 1\ L$$

We can see that by increasing the pressure on a gas, its volume decreases. This is an indirect, or inverse, relationship. As one variable goes up, the other goes down, and vice versa. So, pressure and volume exhibit an indirect relationship.

Now, let's imagine that a gas occupies a volume of 10 mL at a pressure of 40 cm H2O. What would the pressure need to be for the gas to occupy a volume of 22 mL?

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$$P_1V_1 = P_2V_2$$

$$(40 \text{ cm } H2O)(10 \text{ mL}) = (P_2)(22 \text{ mL})$$

$$P_2 = (40 \text{ cm } H2O)(10 \text{ mL})/(22 \text{ mL})$$

$$P_2 = 18.18 \text{ cm } H2O$$

So, let's double-check our logic. According to our answer, in order to increase the volume a gas occupies from 10 mL to 22 mL, we need to reduce the pressure from 40 cm H2O to about 18 cm H2O. Our answer makes sense because we know that pressure and volume have an inverse relationship. In order to increase the volume of a gas, we need to decrease the pressure.

GAY-LUSSAC'S LAW

Gay-Lussac's law describes the direct relationship that exists between pressure and temperature. As you increase the pressure of a gas, it's temperature also goes up and vice versa. So, let's imagine a gas that is at a temperature of 250 degrees Kelvin at a pressure of 120 mmHg. What will the pressure of the gas be if you heat it to 312 degrees Kelvin?

$$P_1T_2 = P_2T_1$$

In this question, P_1 = 120 mmHg, T_1 = 250 degrees Kelvin, T_2 = 312 degrees Kelvin, and we must solve for P_2.

$$(120 \text{ mmHg})(312 \text{ K}) = (P_2)(250 \text{ K})$$

$$P_2 = (120 \text{ mmHg})(312 \text{ K})/(250 \text{ K})$$

$$P_2 = 149.76 \text{ mmHg}$$

So, let's check our logic: Gay-Lussac's law shows a direct relationship between pressure and temperature, meaning that if temperature increases, so does the pressure the gas exerts. In our problem, we increased the temperature. The pressure we calculated in our answer is higher than our starting pressure, so this correlates with what we would expect.

Imagine that we start with a pressure of 2.4 atm and a temperature of 265 Kelvin. What will the temperature of the gas be if we decrease the pressure to 2.1 atm?

In this question, P_1 = 2.4 atm, T_1 = 265 degrees Kelvin, P_2 = 2.1 atm, and we must

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solve for T_2.

$$(2.4 \text{ atm})(T_2) = (2.1 \text{ atm})(265 \text{ K})$$

$$T_2 = (2.1 \text{ atm})(265 \text{ K})/(2.4 \text{ atm})$$

$$T_2 = 231.86 \text{ K}$$

In this example, the pressure decreased. According to Gay-Lussac's law, if the pressure decreases, the temperature should decrease. Our starting temperature was 265 K and our final temperature after the pressure decreased is about 232 K, which is as we would have expected.

Gas Laws Practice Problems

1. What two variables does Boyle's law involve?

2. What two variables does Charles' law involve?

3. What two variables does Gay-Lussac's law involve?

4. A gas exerts a pressure of 23 kPa and has a volume of 2 liters. You want to determine what the pressure would be if it were compressed to 1.5 liters. What gas law would you use?

5. A gas has a temperature of 245 degrees Kelvin at a pressure of 400 cm H2O. You want to determine the pressure it would exert if the temperature were increased to 300 degrees Kelvin. What gas law would you use?

6. A gas has a volume of 30 mL at a temperature of 130 degrees Kelvin. You want to know the temperature it would have if the volume were compressed in half. What gas law would you use?

7. What variable is held constant in Boyle's law?

8. When the temperature of a gas decreases, does the volume increase or decrease?

9. If the volume of a gas increases, does the temperature increase or decrease?

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10. Calculate the drop in temperature when 2.00 L of gas at 300 °K is compressed to 1.1L.

11. What volume change occurs to a 4.0 L gas sample as the temperature increases from 242.0 °K to 300.0 °K?

12. When 60.0 liters of oxygen at 300.0 °K is compressed to 5.00 liters, what must the new temperature be to maintain constant pressure?

13. A gas occupies 1.25 liters at 1.066 atm. What is the pressure if the volume becomes 5.0 L?

14. A gas occupies 45 liters at 3 atm. What is the volume if the pressure becomes 1.0 atm?

15. A gas occupies 2.3 liters at a pressure of 65.0 mm Hg. What is the volume when the pressure is increased to 100.0 mm Hg?

16. A gas occupies 660 liters at a pressure of 1600 psi. What is the volume when the pressure is decreased to 1000.0 psi?

17. Determine the pressure change when a constant volume of gas at 1.50 atm is heated from 220.0 °K to 320.0 °K.

18. Determine the pressure change when a constant volume of gas at 2.7 atm is cooled from 196.0 °K to 120.0 °K.

19. A gas has a pressure of 0.7 atm at 240.0 °K. What is the temperature at standard pressure (1.0 atm)?

20. A gas has a pressure of 312 psi at 299.0 °K. What is the temperature at a pressure of 450 psi?

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Answer Key to Practice Problems

1. Pressure and volume
2. Temperature and volume
3. Pressure and temperature
4. Boyle's Law
5. Gay-Lussac's Law
6. Charles' Law
7. Temperature
8. Decrease
9. Increase
10. 165 °K
11. 4.96 L
12. 25 °K
13. 0.267 atm
14. 135 L
15. 1.495 L
16. 1056 L
17. 2.18 atm
18. 1.65 atm
19. 342 °K
20. 431.25 °K

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